

FANDI ADINATA

Embedded System Engineer

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Kudus, Indonesia

SUMMARY

Embedded System Engineer with over four years of professional experience in electronic hardware design, PCB design, embedded firmware development, and industrial IoT systems. Experienced in designing electronic schematics for analog and digital circuits and multi-layer PCBs, developing embedded firmware for STM32 and ESP32-based systems, and integrating communication protocols including UART, SPI, I²C, RS485 (Modbus RTU), CAN, MQTT, and cellular/Wi-Fi connectivity. Proven experience in developing battery testing equipment, Battery Energy Storage Systems (BESS), industrial automation, and IoT-enabled monitoring solutions, contributing across the full product development lifecycle from hardware design and firmware implementation to system integration, validation, and optimization. Passionate about designing reliable, efficient, and high-quality embedded systems while continuously expanding expertise in electronics, firmware, and product development.

WORK EXPERIENCE

Electrical Engineer R&D

PT. Elektrik Baterai Indonesia (April 2025 – June 2026) – Indonesia

Project: Semi-Automatic Graphene Production System (April 2025 – January 2026)

- Designed electronic schematics and a double-layer STM32-based control board for a graphene production machine, integrating isolated RS485, relay outputs, a 16-bit ADC with an isolated I²C interface for up to 400 VDC voltage sensing and sub-ohm material resistance measurement, and multistage power conversion from 12 V to 5 V and 3.3 V.
- Developed embedded firmware in C for STM32 microcontrollers using STM32CubeIDE, implementing RS485 communication between machine modules.
- Designed and developed the control electronics and embedded firmware for a semi-automatic graphene production system, increasing production capacity from 8.8 g/day to 150 g/day.

Project: 100kWh BESS (March 2026 – April 2026)

- Developed Python software for Modbus RTU data acquisition and MQTT-based telemetry from battery and inverter systems to an internal server for 100 kWh Battery Energy Storage system (BESS).

Project: ELBI Communication Module (February 2026 – June 2026)

- Designed electronic schematics and a double-layer ESP32- and SIMCOM-based communication module for BESS and EVs, supporting 4G LTE cellular and 2.4 GHz Wi-Fi connectivity, and integrating multistage power conversion from up to 150 VDC to 12 V, 4.0 V and 3.3 V within a compact 49.9 mm x 41.9 mm PCB, enabling reliable remote telemetry in deployed systems.
- Optimized embedded firmware for ELBI connection module to acquire data from battery BMS and inverter systems and publish telemetry via MQTT while maintaining power consumption at approximately 0.5 W during data transmission and 0.3 W in idle mode for battery-based system application.
- Improved the thermal performance of the communication module through component placement optimization, reducing maximum temperature rise from 10°C to 6°C (40% reduction) under full-load operation and improving reliability in high-ambient-temperature environments.
- Implemented MQTT communication and automatic cellular/Wi-Fi failover based on user configuration and network availability.

Electrical Engineer R&D

PUI Baterai UNS (January 2025 – April 2025) - Indonesia

- Designed electronic schematics and STM32-based PCB for an 8-channel multi-range battery analyzer (10–100 mA, 0.1–1 A, and 1–5 A at up to 5 V), integrating RS485 communication between the controller and eight battery measurement channels, a 24-bit ADC via SPI communication for precision voltage and current measurement in 0.1 mA resolution with accuracy of $\pm 0.1\%$ FS, and multistage power conversion from 12 V to 5 V, 3.3 V and 2.5 V.
- Developed embedded firmware implementing PID-based charge and discharge control using a 16-bit DAC, achieving current stability of $\pm 1.5\%$ FS at 10–100 mA and $\pm 1.0\%$ FS at 0.1–5 A, with charge/discharge response time <1 second.
- Developed a desktop application using JavaScript, Python, ElectronJS, and PostgreSQL to monitor, control, and acquire real-time data from 8-channel battery analyzer, supporting CC, CV, and CC-CV charge/discharge modes in both single-cycle and continuous cycling operation.

Electrical Engineer R&D

PT. Lectro Energy Semesta (July 2022 – December 2024) - Indonesia

- Designed electronic schematics and STM32-based PCB for DC-DC 72 V LiFePO₄ battery pack charger, integrating high-current buck converter up to 84 V 10 A with efficiency up to 90%, a 16-bit ADC for voltage and shunt-current sensing, an isolated CAN communication to setting output voltage and current, and multistage power conversion from up to 120 V to 12 V, 5 V and 3.3 V.
- Formulated and implemented improvements on performance for the fast-charging-purpose battery pack to achieve high charging current up to 3C.
- Designed electronic schematics and ESP32-based PCB for DC-DC 48 V LiFePO₄ battery pack charger, integrating high-current buck converter up to 54 V 30 A, achieving 80% battery SOC at 26 minutes.

Electrical Staff

PT. Solusi Teknologi Air Nusantara (September 2021 – December 2021) - Indonesia

- Developed the electronic schematics and ATmega2560-based PCB for SIAB's IoT water filtration monitoring system, implementing I²C communication for RTC, integrated with ESP32 module to transmit telemetry over Wi-Fi.
- Developed embedded firmware to perform water turbidity measurement using internal 10-bit ADC, data acquisition, and data transmission to internal server via HTTP connection.

IoT Engineer

PT. Solusi Teknologi Air Nusantara (January 2021 – April 2021) - Indonesia

- Developed the electronic schematics and ATmega2560-based PCB for SIAB's IoT water quality monitoring system, implementing I²C communication for RTC and SPI communication for LCD 5110, integrated with SIMCOM network module to perform remote telemetry in 4G LTE.
- Developed embedded firmware to perform water quality (pH, TDS, and temperature) and flow rate measurement using internal 10-bit ADC, data acquisition, and data transmission to internal server via HTTP connection.

EDUCATION

Universitas Sebelas Maret (UNS) – Surakarta, Indonesia – 2023

- Bachelor's in Electrical Engineering (GPA: 3.58/4.00)

SKILL

Embedded Systems

STM32, ESP32, C/C++

Communication Protocol

UART, SPI, I2C, RS485, Modbus RTU, CAN, MQTT, HTTP

Hardware Design

Double Layer PCB Design, Schematic Design, Power Electronics

Software Development

Python, ElectronJS, PostgreSQL

Development Tools

Git, STM32CubeIDE, Visual Studio Code

Languages

Indonesian (Fluent), English (Intermediate)